OmniFlow Project Report

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1 Project Members

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2 Project Contents

- `omniflow.pdf:` the main research paper of the project. The target submission venue is MobiCom 2027.
- `omniflow-code:` the private source-code repository of the OmniFlow system. It is not open source at this stage.
- `openomnibot:` an open-source Kotlin project component. The repository is available at `https://github.com/omnimind-ai/OpenOmniBot`.

3 Project Introduction

OmniFlow is a GUI Agent cache system for repeated mobile workflows. Modern mobile GUI agents usually rely on large language models or vision-language models to observe a screen, reason about the next action, execute the action, and repeat this loop until a task is complete. This step-wise execution model is flexible, but repeated mobile tasks can pay unnecessary online reasoning cost again and again.

The main idea of OmniFlow is to turn successful GUI executions into reusable `Function` assets. When a similar task appears later, the system first checks whether a matching Function can be reused. If the current GUI state is compatible, OmniFlow executes the cached Function segment directly under control-aware checks. If the cache is missing or unsafe, the system returns to the host agent for normal online reasoning.

The project therefore focuses on practical experience reuse for mobile agents: successful executions are written back as cache assets, future repeated tasks can reuse them, and the host agent remains responsible for new or uncertain states. Most technical details, including system design, implementation, experiments, and limitations, are provided in `omniflow.pdf`.

4 Deliverable Summary

The submitted project material consists of the research paper, the private implementation repository, and the open-source OpenOmniBot component. Together, these artifacts describe a complete project around reducing repeated online reasoning in mobile GUI agents through cache-first Function reuse.